Internal Assessment: Mathematics, Analysis and Approaches

Inversion in Combinatorial Geometry.

Developing the technique and solving problems from the Baltic Way and the IMO Shortlist

Word count: -1

1 Introduction

2 Problem Statements

Problem 1 (Baltic Way 2021)**.** We are given 2021 points on a plane, no three of which are collinear. Among any 5 of these points, at least 4 lie on the same circle. Is it necessarily true that at least 2020 of the points lie on the same circle?

Problem 2 (IMO shortlist 1999)**.** A circle is called a *separator* for a set of five points in a plane if it passes through three of these points, it contains a fourth point in its interior, and the fifth point outside the circle.

Prove that every set of five points such that no three are collinear and no four are concyclic has exactly four separators.

Problem 3 (Baltic Way 2019)**.** There are 2019 points given in the plane. A child wants to draw k (closed) discs in such a manner, that for any two distinct points there exists a disc that contains exactly one of these two points. What is the minimal k, such that for any initial configuration of points it is possible to draw k discs with the above property?

Problem 4 (Baltic Way 2024)**.** There is a set of $N \geq 3$ points in the plane, such that no three of them are collinear. Three points A, B, C in the set are said to form a *Baltic triangle* if no other point in the set lies on the circumcircle of triangle ABC. Assume that there exists at least one Baltic triangle.

Show that there exist at least $\frac{N}{3}$ Baltic Triangles.

3 The Concept of Inversion

Out of nothing I have created a strange new universe. János Bolyai

These problems all involve circles and points, and the simplest solution to each problem uses the common technique of inversion.

3.1 Motivation for Inversion

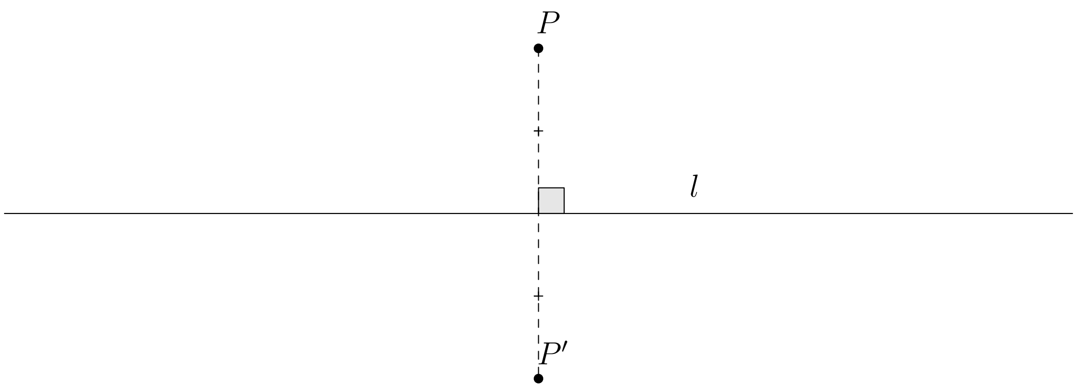

Figure 1: Reflecting a point over a line

Firstly, let us intuitively motivate the upcoming definition of an inversion. We can easily reflect a point across a given line: draw a normal from point onto the line, and mark the refection as the point on this normal which has an equal distance to the line. In **??** point P' is the reflection point P across line l.

Now, what if instead of a line we had a circle instead? How should reflection be defined in that case? This premise may seem odd, but it will turn out to be a surprisingly powerful tool if defined appropriately.

Let us first try to define reflection over a circle in 'the obvious way': Like with the line, draw a normal from the point onto the circle, and mark the reflection as the point on this normal which has an equal distance to the point where the normal intersects the circle. This is quite simple to do, since the normal to the circle from a point is simply the line connecting it with the centre.

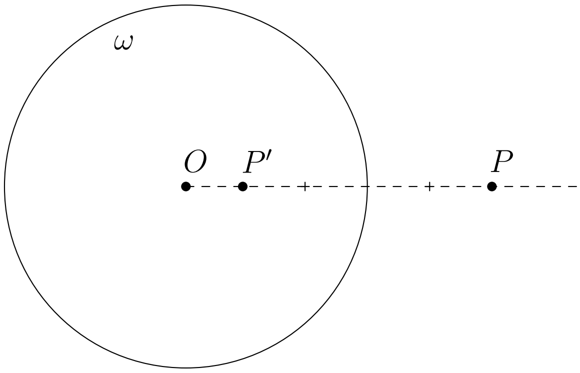

Figure 2: Naïvely reflecting over a circle

In **??**, point O is the centre of circle ω, and line OP is the normal drawn from point P. With this definition, point P' is the reflection of P across ω. Though this looks reasonable so far, upon attempting to make reflections for more points, we realise the drawbacks of such a guileless definition.

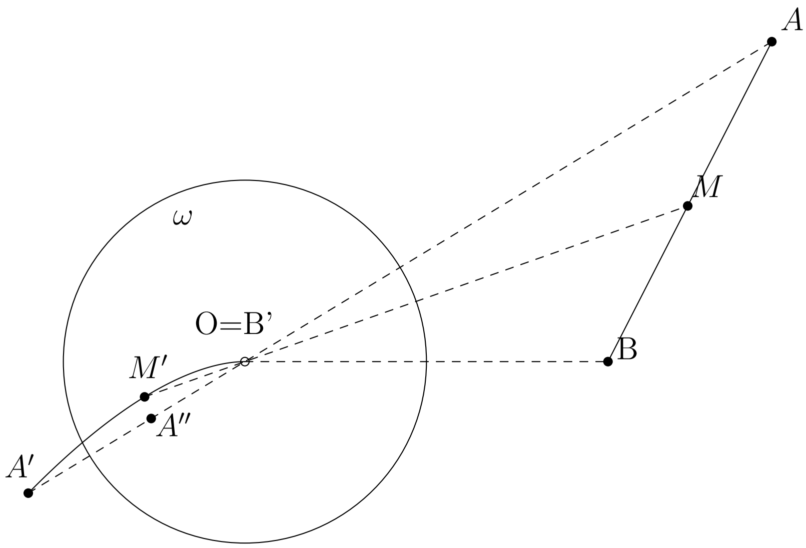

Figure 3: Exposing the weakness of the naïve definition

Again, points A', B' are the reflections of A, B, and likewise A'', B'' are the reflections of A', B'. But making observations from **??** we realise some undesireable implications of this definition.

It feels wrong that A and A' lie on the same side of the circle (the two sides of a circle are 'inside' and 'outside') even though A' is the reflection of A. Moreover, though points A' and B' are the reflections of A and B, points A and B themselves are not the reflections of A' and B'. What even is the reflection of O? It is not possible to define it uniquely since it could be point B, or that point rotated with any angle. Worse still, if point M moves along the segment AB, then its reflection traces some complicated curve with endpoints A', B', instead of something familiar like a line or a circle.

This last issue in particular cripples this definition of reflecting over a circle. When we reflect many points over a line, they still stay the same shape: lines map to lines, circles map to circles; it is not possible for simple things to have complicated reflections.

We need to define the reflection over a circle in a more deliberate way to avoid these undesirable weaknesses. Since we call the operation 'reflection', it should share some of the most fundamental properties fulfilled when reflecting over a line.

Property 1. The reflection of each point exists and is uniquely defined.

Property 2. Reflecting the reflection of a point yields the point itself.

Property 3. Points on the line l are fixed, i.e. reflect to themselves.

Property 4. Points not on the line l are reflected to the opposite side of l.

We can take these properties directly as the requirements for our definition of reflecting over a circle, just by replacing line l with circle ω. We have commented on how the naïve definition fails to satisfy **Properties 1, 2,** and **4**, but **Property 3** is satisfied.

Our note about the reflection of segment AB in **??**, however, warrants another property to be demanded from the upcoming definition:

Property 5. If a point moves along a circle or a line, then its reflection also traces a circle or a line.

Turns out *inversion* is a way to define reflection across a circle which satisfies these properties. In fact, the words are synonymous by convention, and the author apologises for his abuse of terminology in the previous section.

3.2 Definition of Inversion

The author finds no better way to formulate of the definition of inversion than simply taking an extended quotation from his favorite mathematics textbook, *Euclidean Geometry in Mathematical Olympiads* (2016).

> The idea is to view every line as a circle with infinite radius. We add a special point P_∞ to the plane, which every ordinary line passes through (and no circle passes through). This is called the **point at infinity**. Therefore, every choice of three distinct points determine a unique circle or line—three ordinary points determine a circle, while two ordinary points plus the point at infinity determine a line.
>
> With this said, we can now define an inversion. Let ω be a circle with centre O and radius R. We say an **inversion** about ω is a map (that is, a transformation) which does the following.
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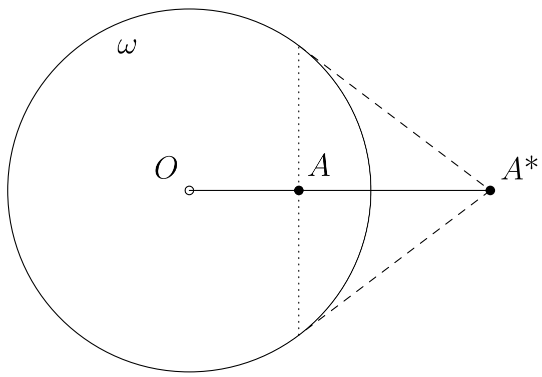
> 
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>
> Figure 4: A^* is the image of point A when we take an inversion about ω
>
> - The centre O of the circle is sent to P_∞
> - The point P_∞ is sent to O.
> - For any other point A, we send A to the point A^* lying on ray OA such that $OA \cdot OA^* = r^2$.
>
> Copied from [1]

3.3 Properties of Inversion

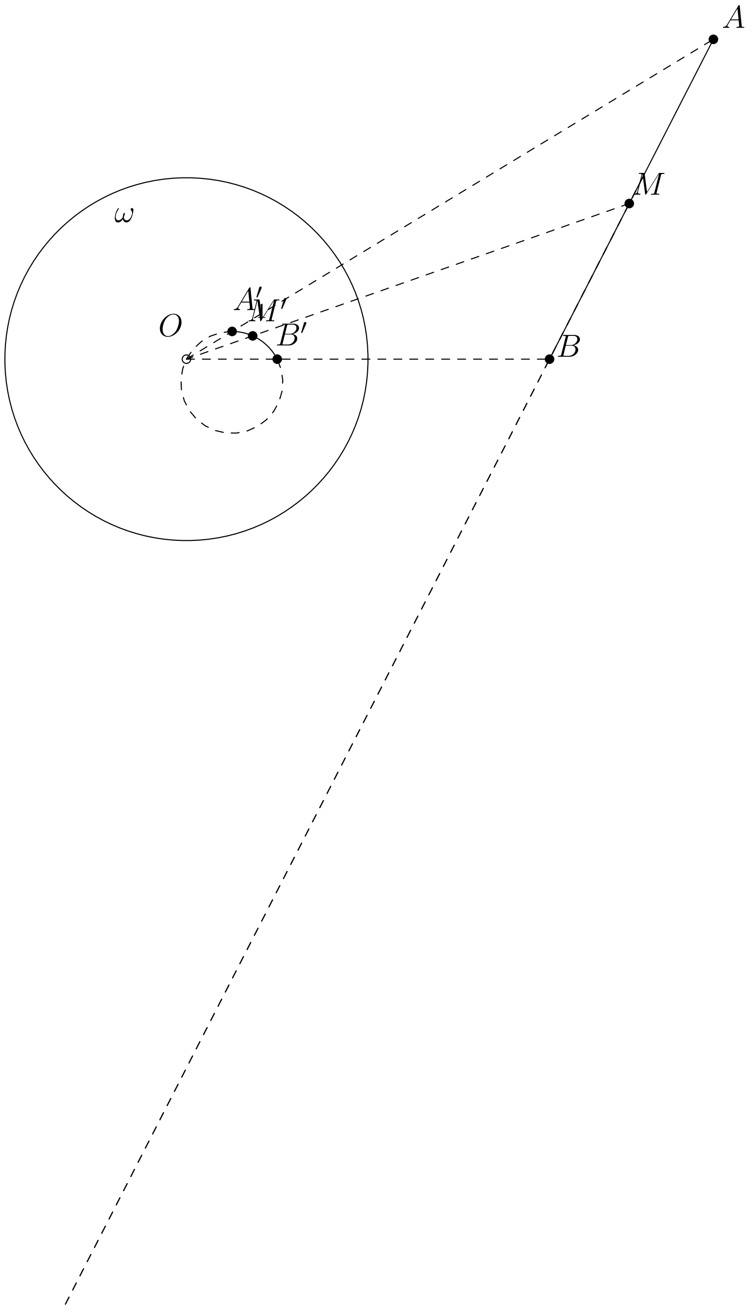
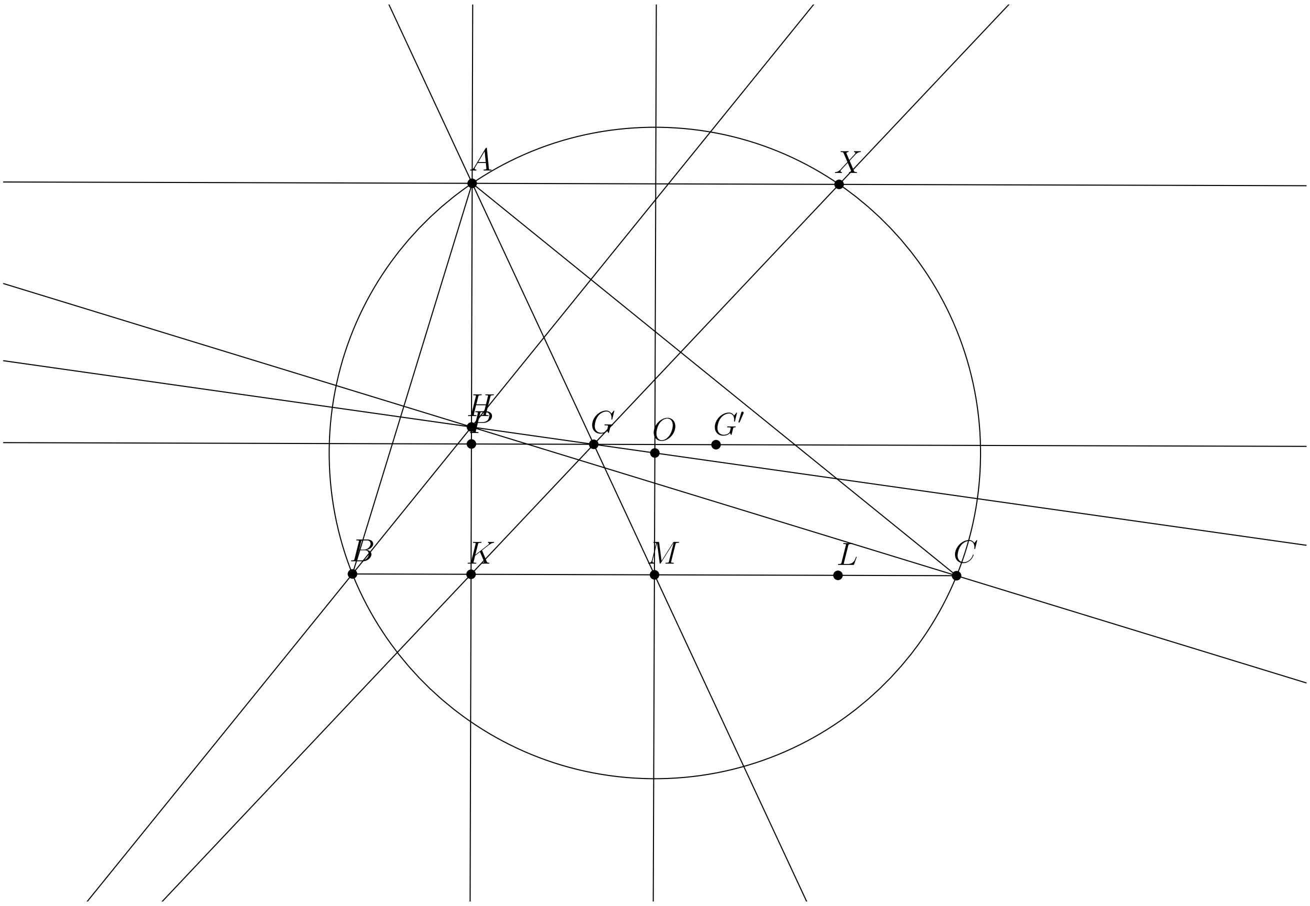


Figure 5: **??** but with inversion

4 Solutions



4.1 Solving Problem 1.

4.2 Solving Problem 2.

4.3 Solving Problem 3.

4.4 Solving Problem 4.

References

[1] Evan Chen. *Euclidean Geometry in Mathematical Olympiads*. MAA Problem Books, 311 pp. Washington, DC: Mathematical Association of America, 2016. ISBN: 9780883858394.